UMA021: Numerical Linear Algebra

L	T	P	Cr
3	0	2	4.0

Course Objectives: The goal of this course is to give students an introduction to numeric and algorithmic techniques used for the solution of a broad range of mathematical problems, with an emphasis on computational issues and linear algebra. In addition, students will become familiar with numeric programming environments Matlab.

Contents:

Roots of Non-Linear Equations: Mathematical preliminaries, bisection, fixed-point, Newton's method and its extension to system of equations.

Interpolation and Integration: Lagrange and Newton basis of polynomials and interpolation problems, divided difference interpolation, forward and backward differences, trapezoidal and Simpson's rules, method of undetermined coefficients.

Matrix Algebra: Gauss elimination method, pivoting strategies, matrix factorization, Jacobi and GaussSeidel methods, matrix norm and conditioning, linear least square problems.

Matrix Computations: Orthogonal and orthonormal basis, Gram-Schmidt process, orthogonal matrices and similarity transformations, power method for eigen-value and eigen-vector, QR algorithm, singular value decomposition.

Laboratory Work:

Lab experiments will be set in consonance with materials covered in the theory and the implementation of numerical methods will be done using MATLAB

Course Learning Outcomes (CLOs) /Course Objectives (COs):

On completion of this course, the students will be able to:

1. Make use of iterative methods to solve nonlinear equations.
2. Approximate the functions using interpolating polynomials and apply to definite integrals.
3. Evaluate solution of system of linear equations and least square problems.
4. Perform matrix computations and evaluate eigen-values and eigen-vectors.

Text Books

1. Richard L. Burden, J. Douglas Faires, and Annette Burden, Numerical Analysis, Cengage Learning, 10th edition, 2015.
2. Gilbert Strang, Linear Algebra and its Applications, Cengage Learning, 4th edition, 2005.
3. J. Desmond Higham and Nicholas J. Higham, MATLAB Guide, Third Edition, Society for Industrial and Applied Mathematics, 2016.

Reference Books

1. Steven C. Chapra and Raymond P. Canale, Numerical Methods for Engineers, McGraw-Hill Higher Education, 6th edition, 2010.
2. E. Ward Cheney and David R. Kincaid, Numerical Mathematics and Computing, Cengage Learning, 7th edition, 2012.

3. Endre Suli and David F. Mayers, An Introduction to Numerical Analysis, Cambridge University Press, 2003